RADHIKA DOTIHAL

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SUMMARY

Software engineer with 5+ years building backend systems for fintech and high-volume data processing. I focus on performance, reliability, and production debugging, with recent work reducing processing time by 83% and speeding up a compute-heavy workflow by 6x. Strong in C++, Java, distributed systems, low-latency architecture, and database design.

EXPERIENCE

Professional Development & Technical Projects **Sep 2025 - Present**

- Completed certifications in AI/LLM systems, Pursued self-paced learning in Terraform, Kafka and Grafana.
- Built a SIMD-optimized matrix multiplication engine using C++, SIMD AVX2 and OpenMP.
- Developed an AI-powered resume analyzer using sentence transformer embeddings and a horizontally-scalable URL shortener.

Software Engineer - SS&C Technologies, Inc. Advent | C++, Python, MySQL, Linux, Containerization, Jenkins, Scrum **Jul 2022 - Aug 2025**

- Reduced report generation time from 18 minutes to under 3 minutes, 83% reduction by redesigning the C++ report engine with multi-threading and batched I/O, eliminating next-day delivery delays and enabling same-day reporting for 5,000+ clients across production.
- Saved 60-70 engineer-hours per month by automating DEV/PROD environment validation, eliminating a 3-hour manual setup process across 20-25 monthly environment builds and reducing onboarding friction for the entire engineering team.
- Reduced environment debugging time by up to 75% by building a Python tool that detected and logged configuration divergence across development runs, cutting diagnosis sessions from 2+ hours to under 40 minutes and eliminating a recurring class of production incidents.
- Collaborated with product, operations, and other cross-functional teams to design scalable backend architecture and integrate new data sources.
- Mentored junior engineers, improved documentation and error-resolution processes, and consistently resolved critical system issues to safeguard data integrity and business continuity.

Volunteer - Global Health Impact Project | Python, Flask, React.js, SQLite **May 2021 - Aug 2021**

- Delivered the backend for a live pharma-impact forecasting platform, still in production to model the global health reach of pharmaceutical products and accelerate evidence-based procurement decisions.

Software Engineer - LTI, India | Hyperion Infra, Oracle EPM, Java, MySQL, HTML, CSS, Bootstrap, JavaScript **Aug 2018 - Nov 2020**

- Reduced release cycle defects by ~80% by orchestrating ETL migration across UAT/DEV/QA environments using Informatica PowerCenter, stabilizing a pipeline that processed 80-110 deployments per quarter and cutting rollback frequency through automated validation gates.
- Migrated OBIEE BI repository serving ~300 users, coordinating across Oracle BI Admin, Enterprise Manager, and Data Integrator, delivering uninterrupted analysis access during a critical platform transition.
- Built a full-stack seat-booking web application (Java,HTML/CSS/BootStrap/JavaScript,MySQL) serving 30-40 users, reducing support ticket volume by 20% through improved input validation, inline error handling and a redesigned booking flow.

PROJECTS

Containers & Mini Docker [Python, Docker, Sysbench, QEMU, GCP]

- Built a Docker-like container runtime, Mini Docker, in Python using Linux namespaces (PID, mount, UTS, network) for process isolation and cgroups for CPU/memory resource limiting.
- Enabled cross-container networking by bridging devices across separate network namespaces using a Linux bridge, allowing isolated containers to communicate - extending the runtime beyond basic isolation to functional inter-container connectivity.
- Compared Mini Docker and Docker against QEMU-based full virtualization, identifying software CPU emulation as key sources of QEMU's higher installation and execution latency.

Live Chat application using Microservices and Orchestration [Python, HTML, JavaScript, MongoDB, Docker, GCP, Microservices]

- Built a simple two-tier chat application starting with the traditional native environment, converting it to microservices setup running with minikube. Created one deployment object for mongoDB, one service object to expose mongodb, one deployment for web server, one service object to expose web server.
- Verified real-time message consistency across multiple concurrent user sessions in the orchestrated microservice deployment, confirming synchronized message delivery using the shared MongoDB backend.

Building & Performance Testing of a Highly Available Website using GCP Serverless Computing [Shell, Docker, JavaScript, Python, GCP]

- Deployed a website to track International Space Station and display location on Google Maps using JavaScript in cloud by utilizing features such as cloud function, cloud SQL and CRON job scheduler.
- Load-tested up to 3,500 concurrent requests using Apache Benchmark, observing GCP autoscaling and adjusting active containers between 13-35 under variable traffic.

Mini Shell [C]

- Developed a shell (console), including system calls like fork, exec, wait, pipe to let the user run the commands with arguments, input and output file redirection and multiple commands separated by pipes.

Matrix Multiplication Optimizer | *C++, SIMD AVX2, OpenMP*

- Engineered a 6x throughput improvement, 0.2 to 1.2 GFLOPS on 1024 X 1024 matrix multiplication by applying cache-blocking, loop-unrolling, AVX2 vectorization and OpenMP parallelism, cutting runtime by 83%, demonstrating C++ performance optimization.

AI Resume Analyzer | Python, Sentence Transformers, Streamlit

- Built a resume screening-tool using the all-MiniLM-L6-v2 that scores resumes against job descriptions using sentence transformer embeddings for semantic similarity, extracting matching and non-matching keywords.
- Designed an interactive Streamlit interface for uploading resumes, job descriptions and visualizing match scores and keyword-level feedback in real time.

EDUCATION

Master of Science, Computer Science @ State University of New York at Binghamton, New York **Feb 2021 - May 2022**

Bachelor of Engineering, Information Technology @ University of Pune, India **Aug 2014 - May 2018**

SKILLS

Languages: C++, Java, Python, JavaScript, HTML5, CSS3

Frameworks: Express.js**,** Flask, JUnit, Pytorch, LangChain

Databases: MySQL, MongoDB, Cassandra, DynamoDB, PostgreSQL

Cloud & Infrastructure as Code: AWS, GCP, Docker, Kubernetes, Terraform, Jenkins

DevOps: CI/CD, Prometheus, Grafana, Kafka

Version Control: Git

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, Design Patterns, System Design, Machine Learning, Operating Systems, DBMS, Web Development, Cloud Computing, REST APIs, Data Science

PUBLICATIONS & ACHIEVEMENTS

- Presented and published research paper titled "***Smart Homes Using Alexa and Power Line Communication in IoT | SpringerLink***" at an annual ICCCNT conference in Springer. **Apr 2018**

CERTIFICATIONS

- Completed online certification in AI Agents, ChatBot Development, LLM fundamentals & AI Ethics. **Apr 2026**
- Pursued self-paced learning in Terraform, Grafana, Apache Kafka 101 and Apache Kafka Connect 101. **May 2026**